

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

25. Perform Sharpening of Image using Gradient masking.

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

AIM:

Sharpening of Image using Gradient masking.

PROGRAM:

```
import cv2
import numpy as np

image_path = r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png"

image = cv2.imread(image_path)

if image is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Compute Sobel gradients in X and Y directions
    gradient_x = cv2.Sobel(gray, ddepth=cv2.CV_64F, dx=1, dy=0, ksize=3)
    gradient_y = cv2.Sobel(gray, ddepth=cv2.CV_64F, dx=0, dy=1, ksize=3)

    # Calculate the gradient image
    gradient = cv2.subtract(gradient_x, gradient_y)

    # Convert to 8-bit for proper display and saving
    gradient = cv2.convertScaleAbs(gradient)

    # Save the output image
    cv2.imwrite(
        r"C:\Users\Susmitha\Downloads\ITA0510 Lab\sharpened_image3.jpg",
        gradient
    )
```

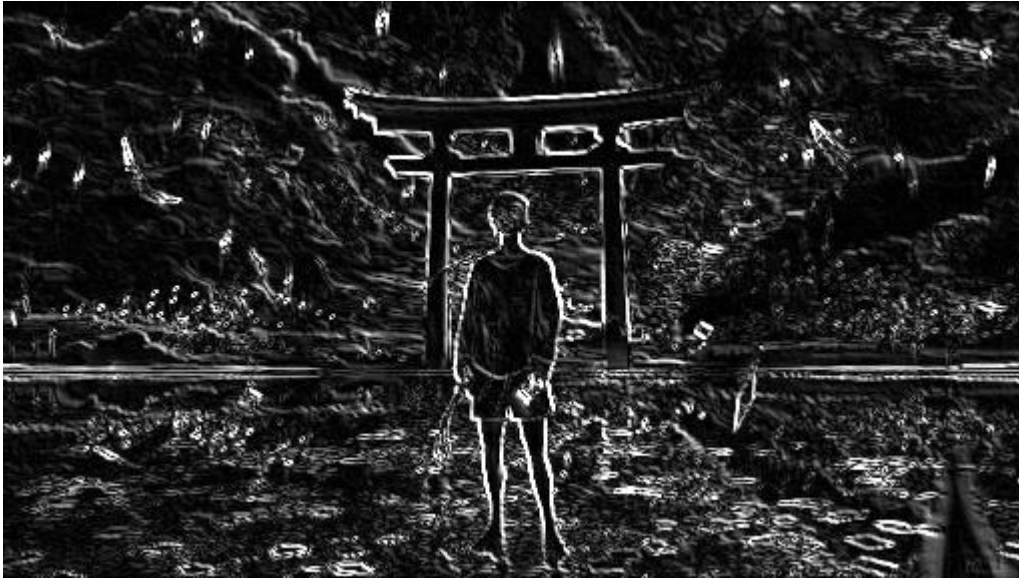
```
# Display the images
cv2.imshow("Original Image", image)
cv2.imshow("Gradient Image", gradient)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.